

Part 1 - Data Analytics Insights Report

Smart City Traffic Intelligence: From Data Analytics to AI-Powered Mobility

Overview

This analysis examined historical traffic patterns on westbound I-94 near Minneapolis-St Paul using SQL, descriptive statistics, probability analysis, and Power BI visualisation. The objective was to identify meaningful traffic patterns, assess how weather and temperature relate to congestion, and translate the findings into practical insights for a smart-city mobility team.

Key Findings

Finding	Summary
Annual traffic trends	Year-to-year totals require caution because 2012, 2014, and 2015 have incomplete coverage. The cleanest full-year comparison is 2016 to 2017, when total traffic increased from 29,471,608 to 35,393,801 vehicles, an increase of approximately 20.09%.
Traffic variability	The mean traffic volume is 3,259.62 vehicles and the median is 3,379. The standard deviation is 1,986.95 vehicles and the overall range is 7,280 vehicles, showing substantial variation across hourly records.
Temperature relationship	The Pearson correlation coefficient between temperature and traffic volume is 0.1323, indicating a very weak positive relationship. The Power BI scatter plot also shows no clear visual relationship: traffic levels are broadly similar across approximately -20 C to +30 C, with no distinct temperature band associated with higher traffic.
Congestion probability	Congestion, defined as traffic volume above 5,500 vehicles, occurred in approximately 14.73% of records. Clear weather occurred in 27.78% of records, while 3.66% of all records had both clear weather and congestion. Among congested records, 24.83% occurred during clear weather and 26.30% occurred when temperature exceeded 292 K.
Weather and congestion	$P(\text{Clear AND Congestion}) = 0.0366$, compared with $P(\text{Clear}) \times P(\text{Congestion}) = 0.0409$, suggesting clear weather and congestion are not perfectly independent. The congestion odds ratio for clear versus cloudy weather is 0.7355, so the observed odds of congestion in clear weather were about 26% lower than in cloudy weather.
Hourly pattern	The 2017 Power BI analysis shows traffic rising sharply during the morning commute and remaining relatively high through the day, with the highest average traffic occurring around 4 PM.
Weather-condition comparison	Clouds had the highest average traffic volume at 3,618.45 vehicles, while Squall had the lowest at 2,061.45 vehicles. The difference between the highest and lowest average traffic was 1,557 vehicles.

Data Quality and Limitations

The Power BI dashboard is based on the raw dataset containing 48,204 rows. The Python/SQLite cleaned dataset contains 48,187 rows after removing 17 exact duplicate records. This difference is intentional and should be kept in mind when comparing totals or counts between the Power BI dashboard and the cleaned analytical outputs.

Power Query showed no errors or empty values across the raw 48,204-row dataset. However, the analysis identified ten invalid temperature readings recorded as 0 K. For the Power BI Celsius field, these invalid records were set to null before conversion so that they would not distort the temperature analysis. The earlier-year coverage gaps also limit direct interpretation of some annual totals.

Implications for the Smart-City Mobility Team

The strongest operational signal is the time-of-day pattern rather than temperature. Traffic-management resources should therefore focus on predictable commuting peaks, particularly the late-afternoon period around 4 PM. Adaptive traffic signalling, congestion alerts, travel-time messaging, and public-transport capacity planning could be concentrated around known high-demand periods.

Weather can still be used as a secondary contextual factor. Cloudy conditions were associated with higher average traffic and clear weather with somewhat lower congestion odds, but the weak temperature correlation shows that weather alone is not sufficient for predicting traffic conditions.

The incomplete coverage in several early years is also important. Long-term trend decisions should rely mainly on complete data periods, and future data collection should aim for consistent coverage so that year-on-year comparisons remain reliable.

Conclusion

Westbound I-94 traffic is highly variable, with clear daily and hourly demand patterns. Congestion occurs in about 15% of observations and is associated with multiple factors. Temperature has little direct relationship with traffic volume, while time-of-day and commuting behaviour provide stronger operational signals. For the smart-city mobility team, the analysis therefore supports prioritising time-based congestion management while using weather as a supporting, rather than primary, decision variable.

Recommended dashboard use: Use the Power BI hour-range, weather-condition, and traffic-category slicers to explore how the key patterns change under different operating conditions. The dashboard should be treated as an interactive decision-support tool rather than a causal model.